

The Bounded-Torsion Sidon Decomposition Problem Is Answered

Literature-status note for arXiv:1607.01053

19 July 2026

Status. Literature already answered, partial subcase. Lewko’s arXiv:2606.06669 answers the bounded-torsion, and hence exponent-4, case of Pisier’s decomposition question. The general conjecture remains open.

Original question

Gilles Pisier, *Subgaussian sequences in probability and Fourier analysis*, arXiv:1607.01053v3 (2016), formulates in Section 7 the conjecture that every subgaussian set of characters on a compact Abelian group is a finite union of quasi-independent sets [1]. The survey explains the equivalence with the corresponding Sidon-set problem.

After recalling the prime-exponent theorem and Bourgain’s squarefree-exponent extension, Remark 7.3 on source PDF page 23 states that the equivalence was still open even for exponent $p = 4$.

Answering theorem

Lewko’s 2026 paper *The Sidon Decomposition Problem in Abelian Groups of Bounded Torsion* proves the needed result. Theorem 1.2, on supporting PDF page 2, states [2]:

If the discrete dual group $\Gamma = \widehat{G}$ has bounded torsion, then a set $\Lambda \subseteq \Gamma \setminus \{0\}$ is Sidon if and only if it is a finite union of quasi-independent sets.

Taking $\Gamma = (\mathbb{Z}/4\mathbb{Z})^I$ directly answers Pisier’s highlighted exponent-4 subproblem. More generally, the theorem resolves every fixed bounded-torsion case. This is an explicit literature answer rather than a new proof produced by the present run: Lewko cites Pisier’s survey, describes the prime-power obstruction, and says that the main theorem answers the outstanding bounded-torsion question.

Scope limitation

The original conjecture ranges over arbitrary compact Abelian groups. Lewko’s paper explicitly notes that the analogous problem for $\Gamma = \mathbb{Z}$ remains open. Thus the status is a complete answer for bounded torsion, but only a partial subcase of Pisier’s general question.

Search record

The run’s registry, solution, attempt, and proof-gap indexes were searched for arXiv:1607.01053, arXiv:2606.06669, and subgaussian/Sidon/quasi-independent decomposition terms. No prior packet

was found. A targeted arXiv search on 19 July 2026 located the June 2026 supporting paper. Both official arXiv PDFs are stored with this note.

References

- [1] G. Pisier, *Subgaussian sequences in probability and Fourier analysis*, Graduate J. Math. 1 (2016), 59–78; arXiv:1607.01053v3.
- [2] M. Lewko, *The Sidon Decomposition Problem in Abelian Groups of Bounded Torsion*, arXiv:2606.06669, 2026.